Computer Security

Euler's-totient Function:

$\phi(n)$ for $[n \geq 1]$ is defined as the number of the integers less than 'n' and are Coprime to 'n'. [ie: $gcd(n,x)=1$]

e.g

$$\phi(5) = \{1,2,3,4\} = 4$$

$$\phi(6) = \{1,5\} = 2$$

⇒ When 'n' is a Prime number,

$$\phi(n) = n-1 \quad [\phi(23) = 22]$$

⇒ $\phi(a*b) = \phi(a)*\phi(b)$ [a & b are Coprime]

$$\phi(35) = \phi(7*5) = \phi(7)*\phi(5)$$

$$= 6*4 = 24$$

Euler's Theorem:

If $gcd(x,n)=1$ then,

$$\boxed{x^{\phi(n)} = 1 \bmod n}$$

e.g $x = 4$ and $n = 165$

So, $4^{\phi(165)} = 1 \bmod 165$ [as $gcd(4,165) = 1$]

Now,

$$\phi(165) = \phi(15) \times \phi(11)$$
$$= \phi(3) \times \phi(5) \times \phi(11)$$
$$= 2 \times 4 \times 10 = 80$$

$$\Rightarrow 4^{80} = 1 \bmod 165$$

Else,
$$\boxed{x^{\phi(n)\,a} = 1 \bmod n}$$ [a = any Integer]

Fermat's Theorem:

From Euler's,

$$x^{\phi(n)} = 1 \bmod n$$

If [n = Prime] $\Rightarrow \phi(n) = n-1$

$$\Rightarrow \boxed{x^{n-1} = 1 \bmod n}$$

n = Prime
'x' is not divisible by 'n'
$(x \not\equiv 0 \bmod n)$

e.g $x = 3$ and $n = 5$

$$3^{5-1} = 3^4 = 81$$

$$\Rightarrow 81 = 1 \bmod 5$$

Now, $x^{\phi(n)} = 1 \bmod n$ → Euler's Theorem

$\Rightarrow x^{n-1} = 1 \bmod n$ → Fermat's

$\Rightarrow x \cdot x^{n-1} = x\,(1 \bmod n)$

$\Rightarrow x^{n-1+1} = x \bmod n$

$\Rightarrow \boxed{x^{n} = x \bmod n}$ → n → Prime

$\gcd(n, x) = 1$

RSA Algorithm (Rivest, Shamir, Adlemann)

Step ①: Choose 2 Prime ⇒ $P = 61$ & $q = 53$

Step ②: Compute $P \times q = n$

$\Rightarrow n = 61 \times 53 = \boxed{3233}$

Step ③: $\phi(n) = \phi(3233) = \phi(61 \times 53)$

$= \phi(61-1) \times \phi(53-1)$

$= 60 \times 52 = \boxed{3120}$

Step ④: Choose 'e' [i.e $1 \leq e \leq \phi(n)$ and $\gcd(e, \phi(n)) = 1$]

$\Rightarrow e = 17$

∴ Public key ⇒ (e, n) ⇒ $(17, 3233)$

Step(5) : Determine 'd' as, $[ed = 1 \bmod \phi(n)]$

$$\Rightarrow d = e^{-1} \bmod \phi(n)$$

Now,

$$17 \times d = 1 \bmod 3120$$

$$\Rightarrow d = 2753$$

$$d = \frac{(\phi(n) * i) + 1}{e} = \frac{(3120 * i) + 1}{17}$$

$[i = 1, 2, 3, 4 \text{------}]$

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[Continue to calculate this eq^n untill 'd' = Integer]

$$\boxed{\therefore \text{Private key} \Rightarrow (d, n) = (2753, 3233)}$$

For Encryption.

$\boxed{P < n}$

$$\boxed{C = P^e \bmod n}$$

Decryption

$$\boxed{P = C^d \bmod n}$$

P → Plain text

Public key (e, n) } will be fixed
Private key → (d, n) }

C → Cypher text

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CIA Triad:

(i) Confidentiality: Prevent or minimize unauthorized access of data (like- encryption)

(ii) Integrity: Protecting the reliability and Correctness of data, unauthorized alterations of data (like- hashing, Checksum)

(iii) Availability: Ensuring that systems and information are accessible when needed.

Diffie Hellman Key Exchange:

Alice

(i) Public key = P, G
(ii) Private Key = a
(iii) Key generation

$$\boxed{x = G^a \bmod P}$$

Bob

(i) Public Key = P, G
(ii) Private Key = b
(iii) Key Generation,

$$\boxed{y = G^b \bmod P}$$

(iv) Exchange of Key [$x \leftrightarrow y$]

Alice

(v) Receiving key = y
(vi) Generate secret key,

$$\boxed{K_a = y^a \bmod P}$$

Bob

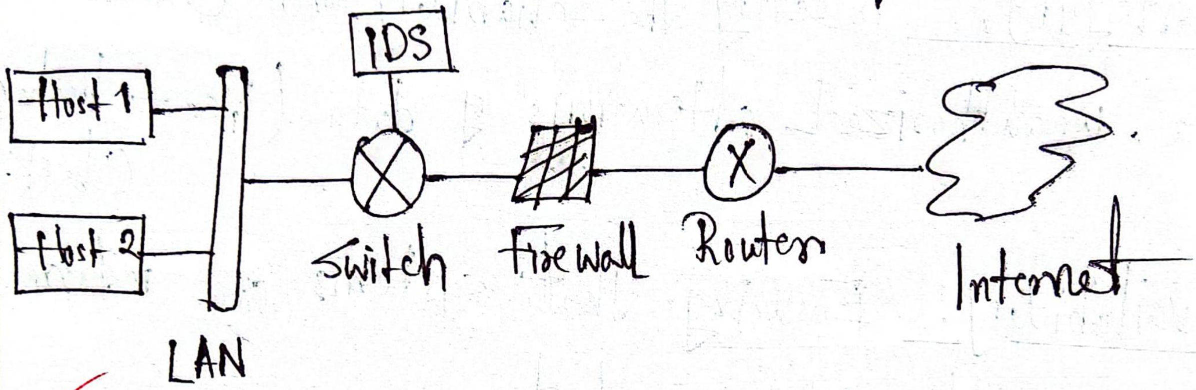
(v) Received key = x
(vi) Generate Secret key,

$$\boxed{K_b = x^b \bmod P}$$

(vii) Algebrically, $\boxed{K_a = K_b}$

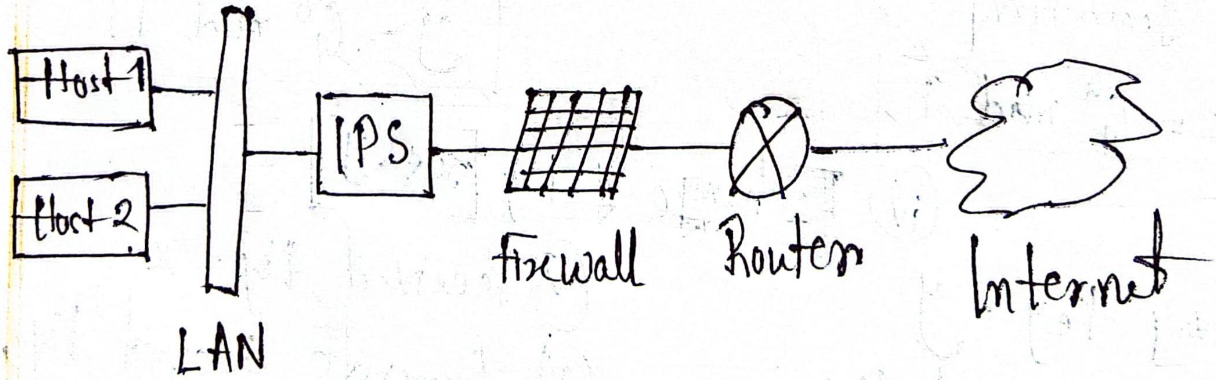
Intrusion Detection System (IDS)

⇒ It's a device or software application that monitors a traffic for malicious activity or Policy violation and sends alert on detection.



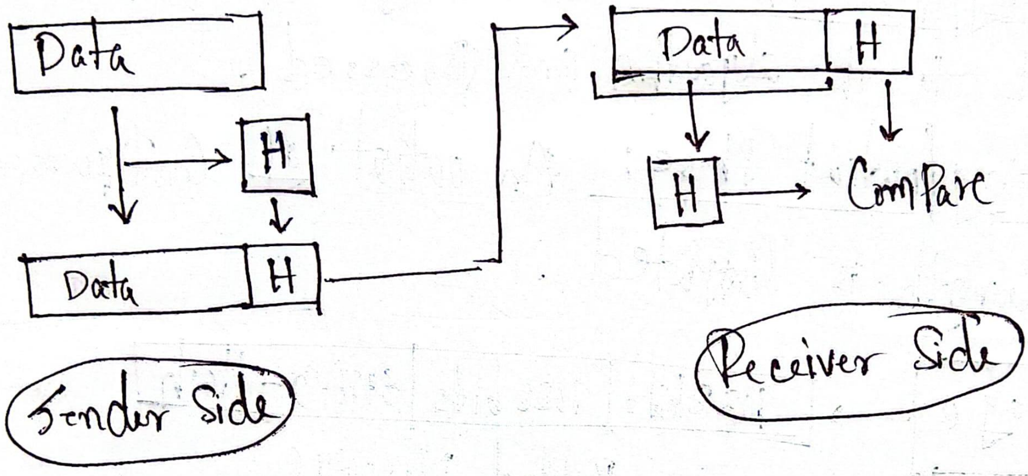
Intrusion Prevention System (IPS)

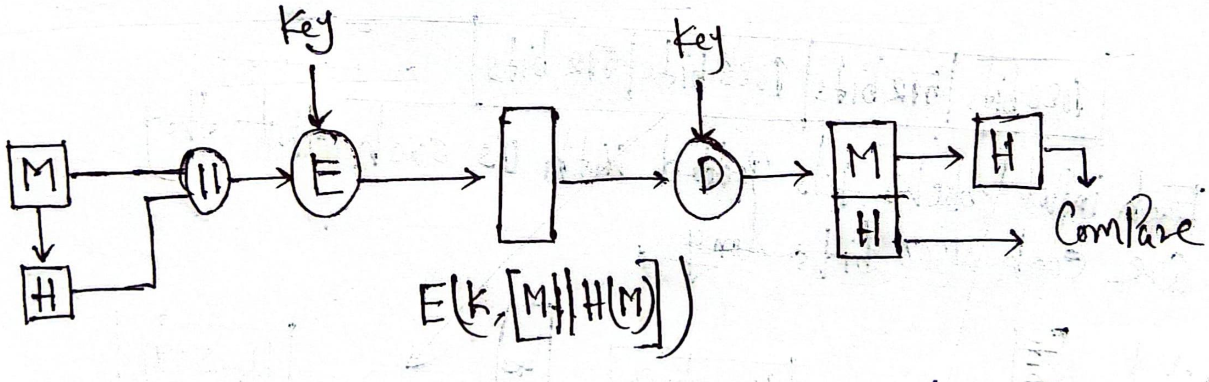
⇒ Inspects real time traffic and looks for traffic patterns or signatures of attack and then Prevents the attacks on detection.



Hash Function :

→ The Principal object of a hash function is data integrity. It's often used to determine whether or not data has changed.





Messege → Hash Message → Encryption ⇒ Decryption → Message + Hash

Sender side

→ Hash Conversion of Message

→ Compare with sender hash

Receiver side

(i) Sponge Construction (ii) Absorbing (iii) Squeezing

Secure Hashing Algorithm-3 (SHA-3)

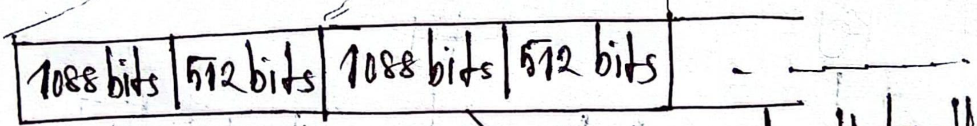
⇒ This hash function is based on Sponge Construction - Consist of 2 Phases →

(i) Absorbing/Input Phase: The message block x_i are Passed to the algorithm and Processed.

(ii) Squeezing/output Phase: An output of Configurable length is Computed.

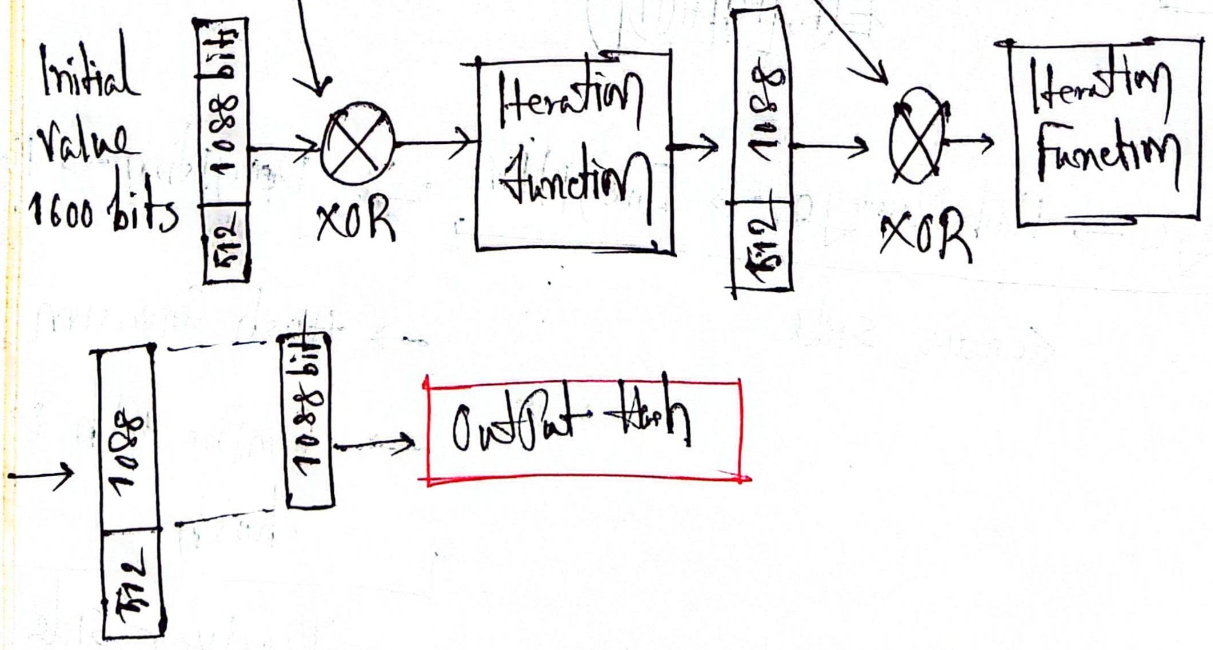
Step (1)

Step (2)



Each input blocks gets Padded with 0's such that they are each 1600 bits long.

Step (3)



Message Authentication Requirement

⇒ It is a Procedure to verify that received messages from the source and have not been altered.

It also verify sequencing and timeliness.

⇒ While transmitting message accross a network, the following attacks Can be ~~ing~~ identified —

(i) Content modification

(ii) Sequence "

(iii) Timing "

(iv) Source Repudiation

(v) Destination "

Digital Signature :

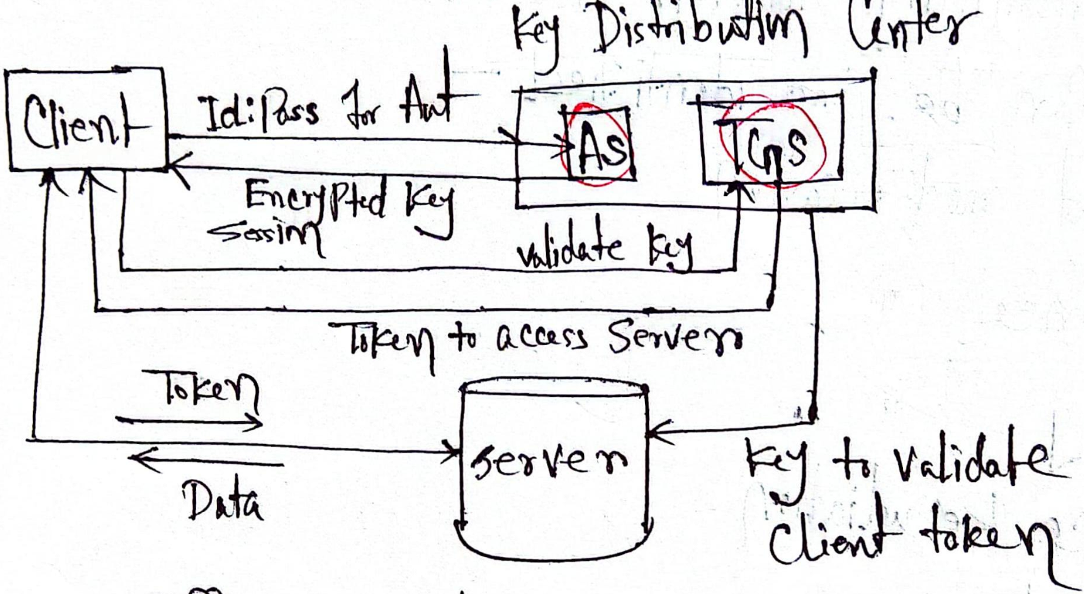
It is an authentication technique that also includes measures to Counter repudiation by the source.

[Diagram refers to the Previous "Hash Function's" diagram]

The Converted message hash $\boxed{H}$ is Called the digital signature here.

Kerberos:

⇒ It's a authentication Protocol that authinticate the ~~se~~ actual sender and receiver in the net-work to tranimit data.



step ① : C → As

② : As → C

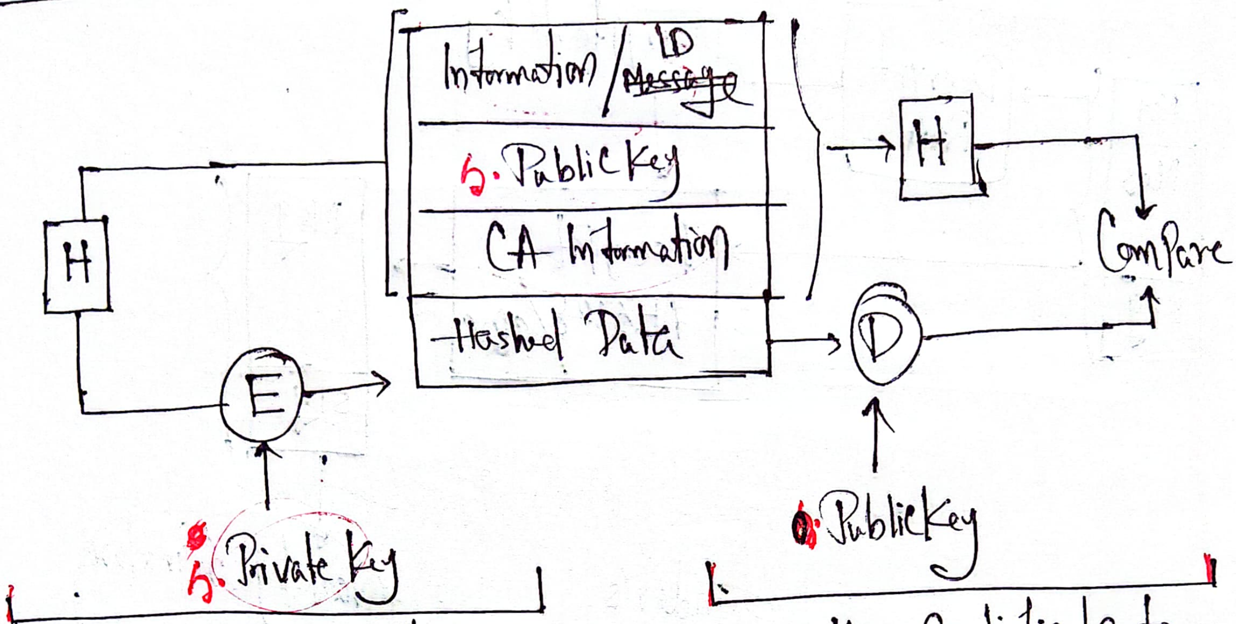
③ : C → V

C → Client

As → Auth. Server

V → Server

X.509 Certificate:



Create signed digital Certificate

Use Certificate to verify information / ID

S/MIME: Secure/Multipurpose Internet Mail Extension

⇒ It is a security enhancement to the MIME internet mail format standard based on technology from RSA data security.

⇒ Most commonly use RSA and SHA-256

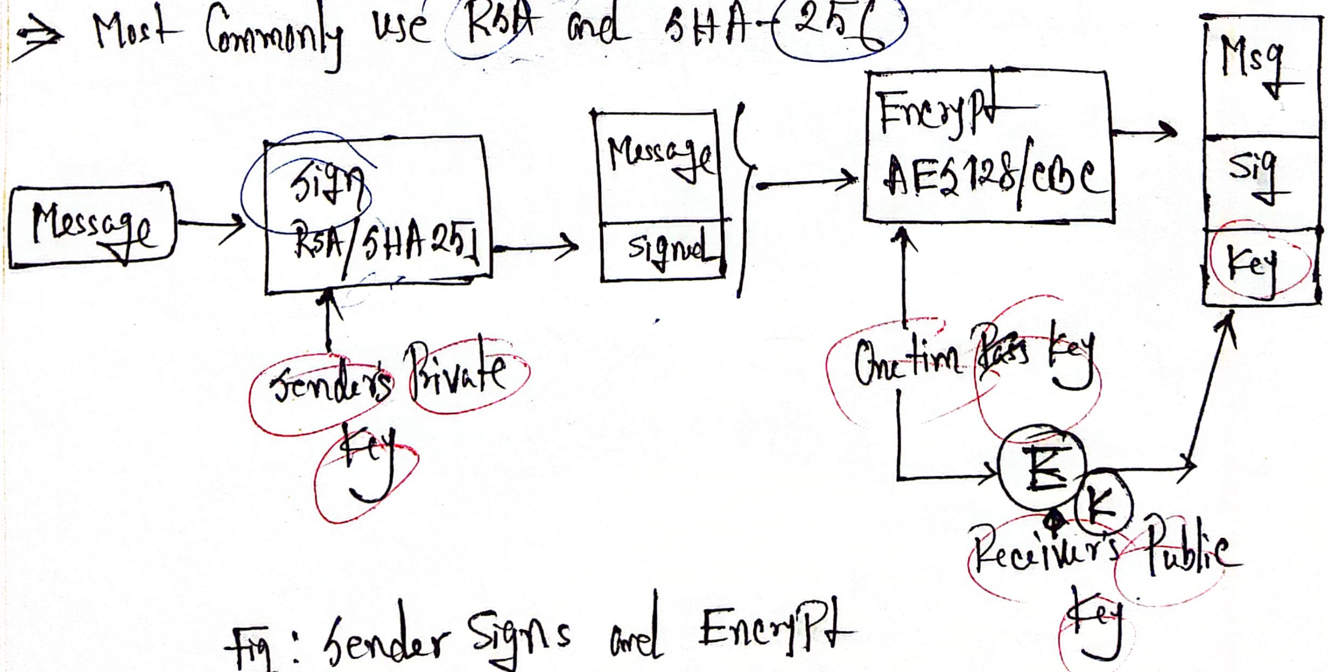


Fig: Sender Signs and Encrypt Message.

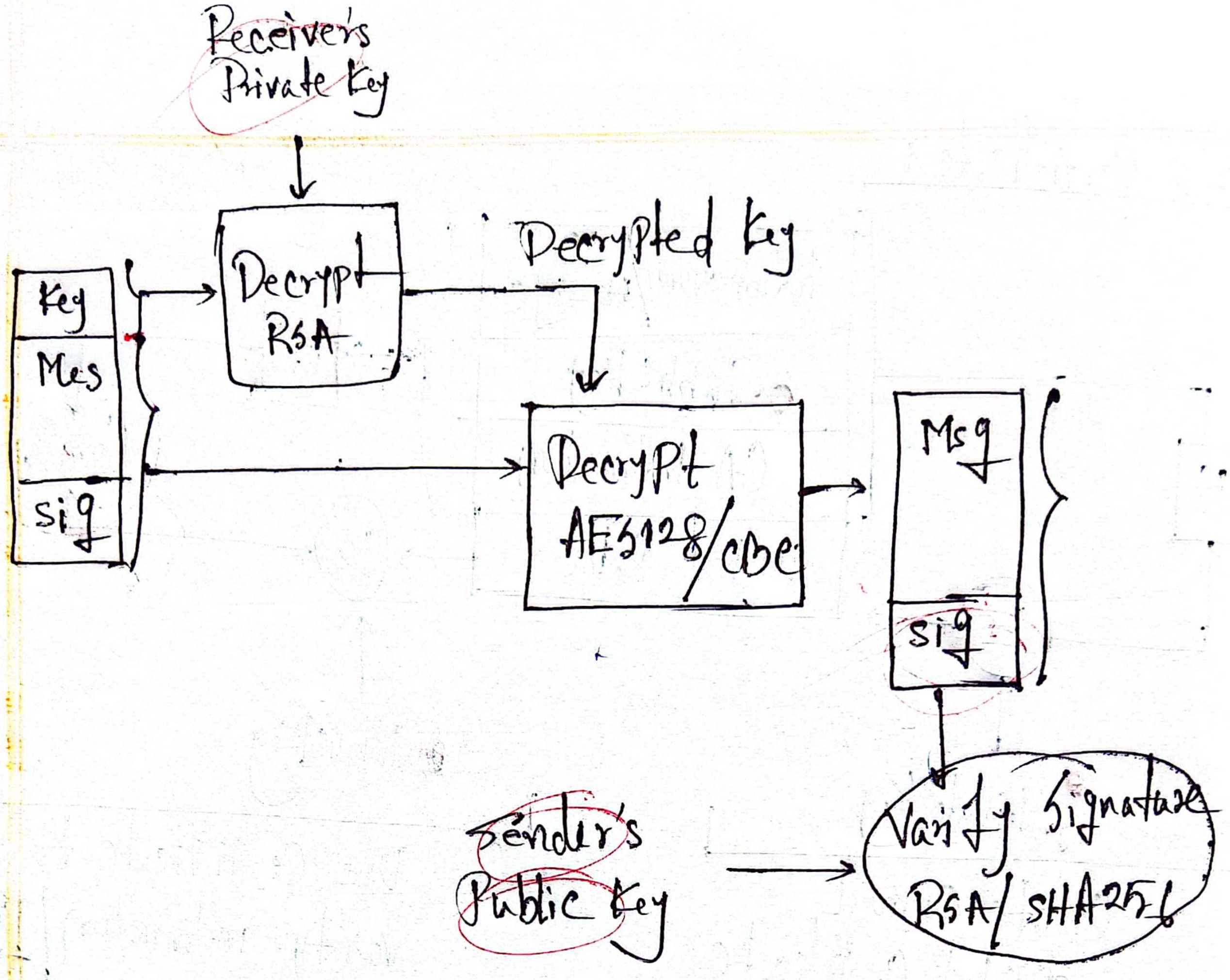


Fig: Receiver decrypts msg and varifies with Sender's Public Key.

Segment-6

IP Level Security:

By implementing security at IP level, an organization can essure secure networking not only for applications that have security mechanisms but also for the many security ignorant applications.

⇒ 3 Functional areas —

(i) Authentication (ii) Confidentiality (iii) Key management.

IPSEC has two modes —

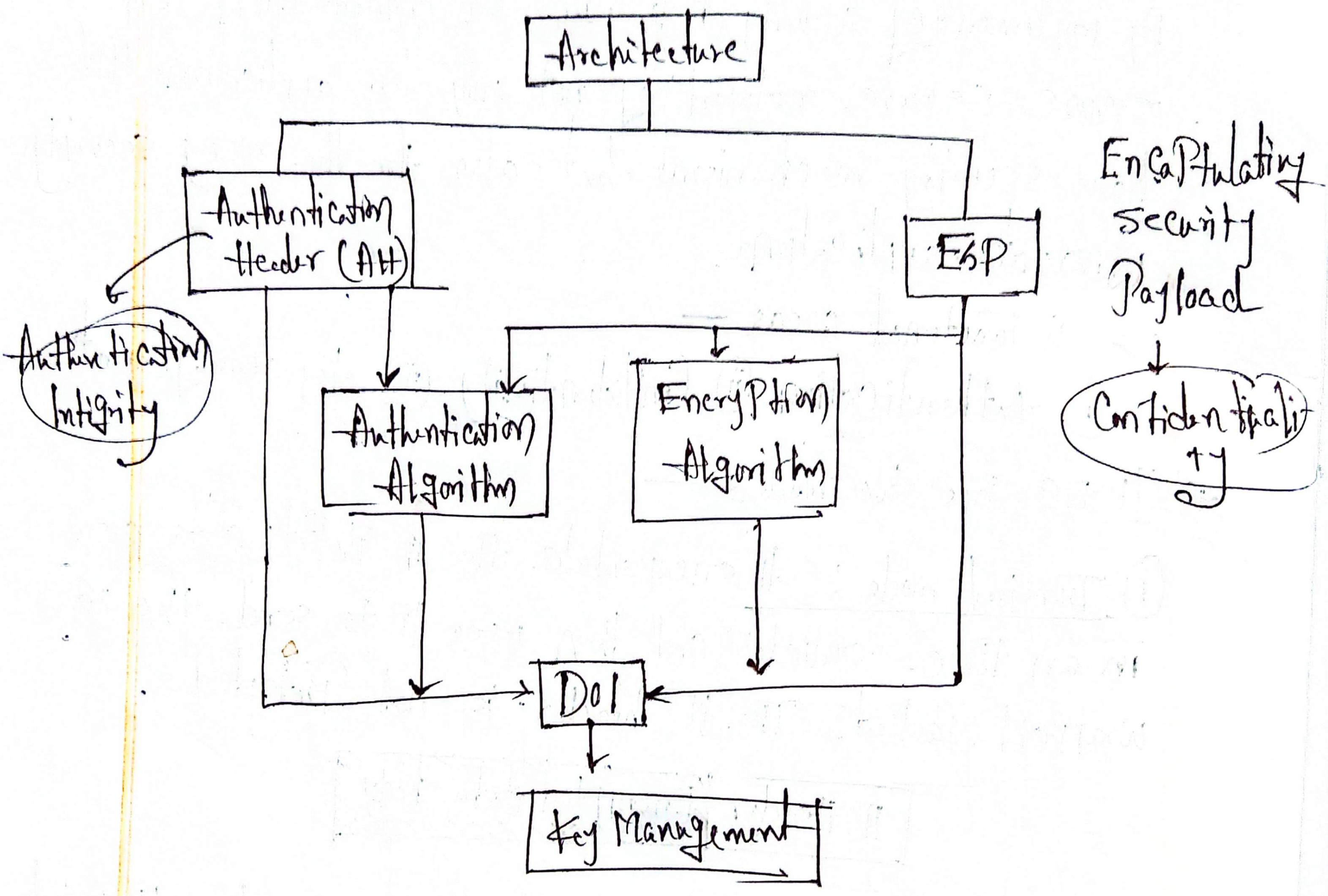
(i) Transport mode: It encapsulate the IP Packet data area in an IPsec envelop and then uses IP to send the wrapped Packet. The IP header is not Protected.

IP Header	Encrypted data body

(ii) Tunnel mode: It encapsulate the entire IP Packet with data and then forward it using IP. Here the IP header of the encapsulated Packet is Protected.

IP Header	Encrypted IP Header	Encrypted Data body

IP security Architecture :



Encapsulating Security Payload (ESP) :

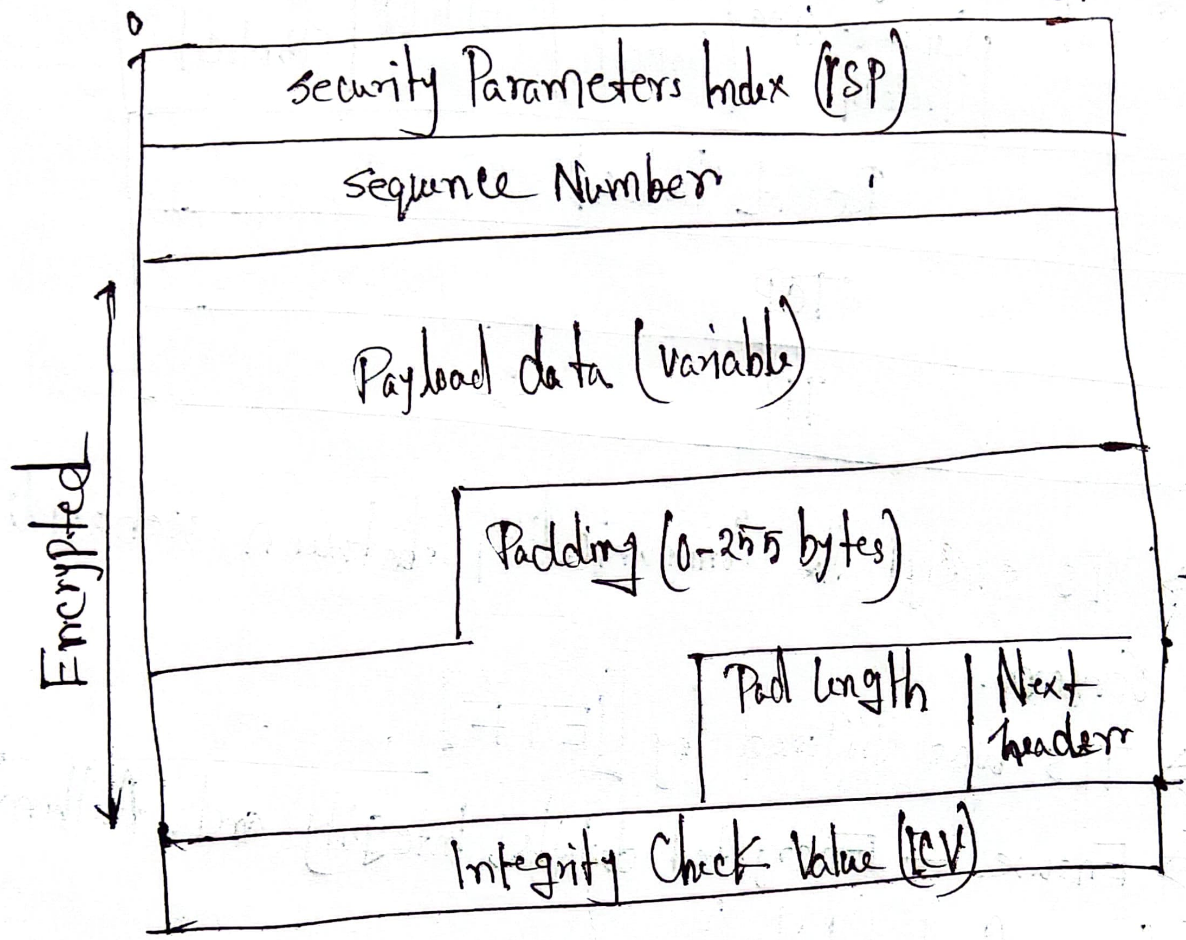


Fig: Top level format of an ESP Packet

Transport layer Security:

⇒ TLS is designed to make use of TCP to Provide a reliable end-to-end secure service.

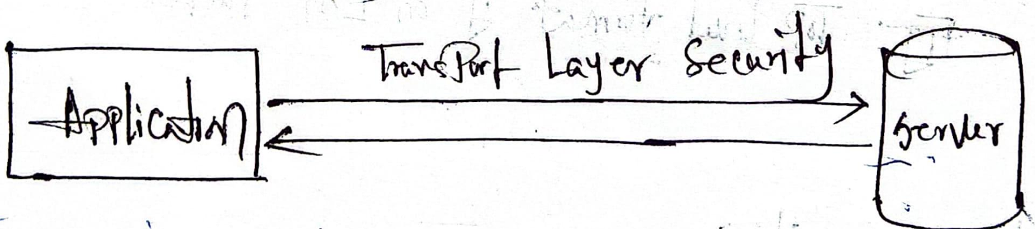
Related to Secure Socket Layer (SSL) and the HTTP is based on top of TLS.

<table>
<tr><td>Handshake Protocol</td><td>Change Cipher Spec Protocol</td><td>Alert Protocol</td><td>HTTP</td><td>Heartbeat Protocol</td></tr>
<tr><td colspan="5">Record Protocol</td></tr>
<tr><td colspan="5">TCP</td></tr>
<tr><td colspan="5">IP</td></tr>
</table>

⇒ TLS encryt the Communication between webapplication and Serverr.

⇒ TLS was Proposed by [IETF].

⇒ Ensure — Encrypted details, Integrity and Authentication (e.g: Card details)



* ⇒ Overcome drawbacks of SSL.

Wireless Security Challenges

1. Channel: Dropping and Jamming.
2. Mobility:
3. Resources — Limited memory and Processing speed
4. Accessibility

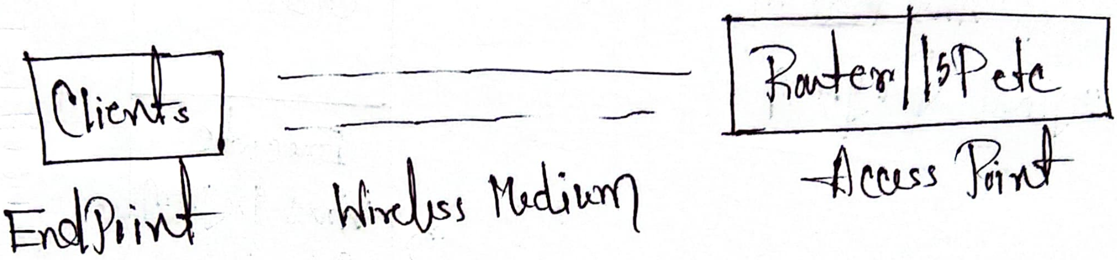


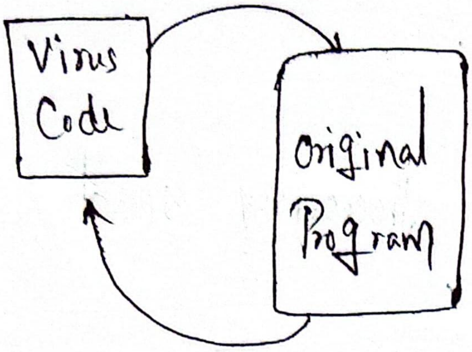
Fig: Wireless Network Components.

Wireless Network Threats:

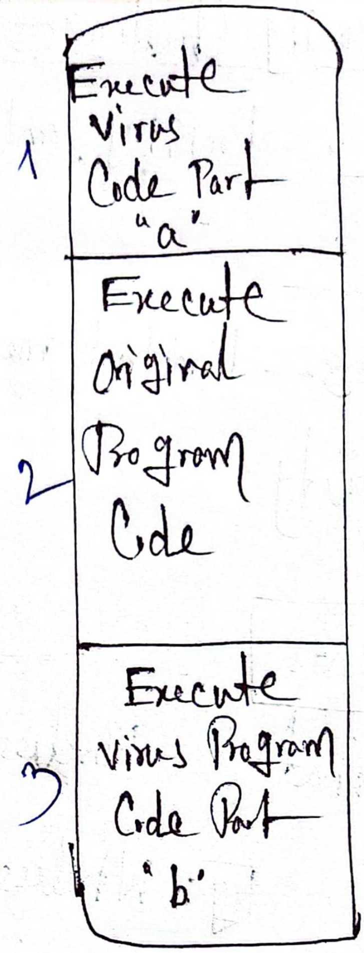
1. Malicious association
2. Ad-hoc network → In Peer-to-Peer network no middle Control point.
3. Identity theft → attackers is able to identify the MAC address of target device.

Zombie: It's a Program that secretly takes over another Internet-attached Computer and then uses that Computer to launch attack to other target device.

Virus Structure:



Push virus Code in original Program



Execution of virus through a Program file.

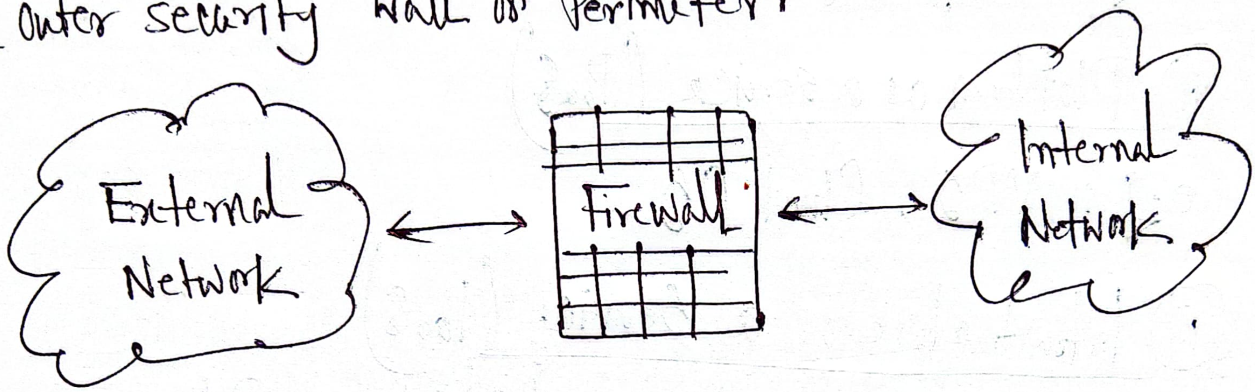
Intruders: (Hacker/Cracker)

3 types of intruders —

(i) Masqueraders
(ii) Misfeasor
(iii) Clandestine User

Firewall :

⇒ The firewall is inserted between the Premises network and the internet to establish a Controlled link and to build an outer security wall or Perimeter



Types of firewall :

(i) Packet filters
(ii) Application level gateway / Proxy server
(iii) Circuit Level Gateway

Cloud Computing :

⇒ It is the delivery of Computing services over the internet such as servers, storage, database, networking software, analytics and intelligence.

Types of Cloud Computing Services :

(i) Software as a service (SaaS)

e.g : Gmail - google email service

(ii) Platform as a service (PaaS)

e.g : VPS , Cloud OS

(iii) Infrastructure as a Service (IaaS)

e.g : Data storage , highly adaptable Computer System.

4 Phases of Virus Lifetime :

(i) Dormant Phase [Access target Computer]

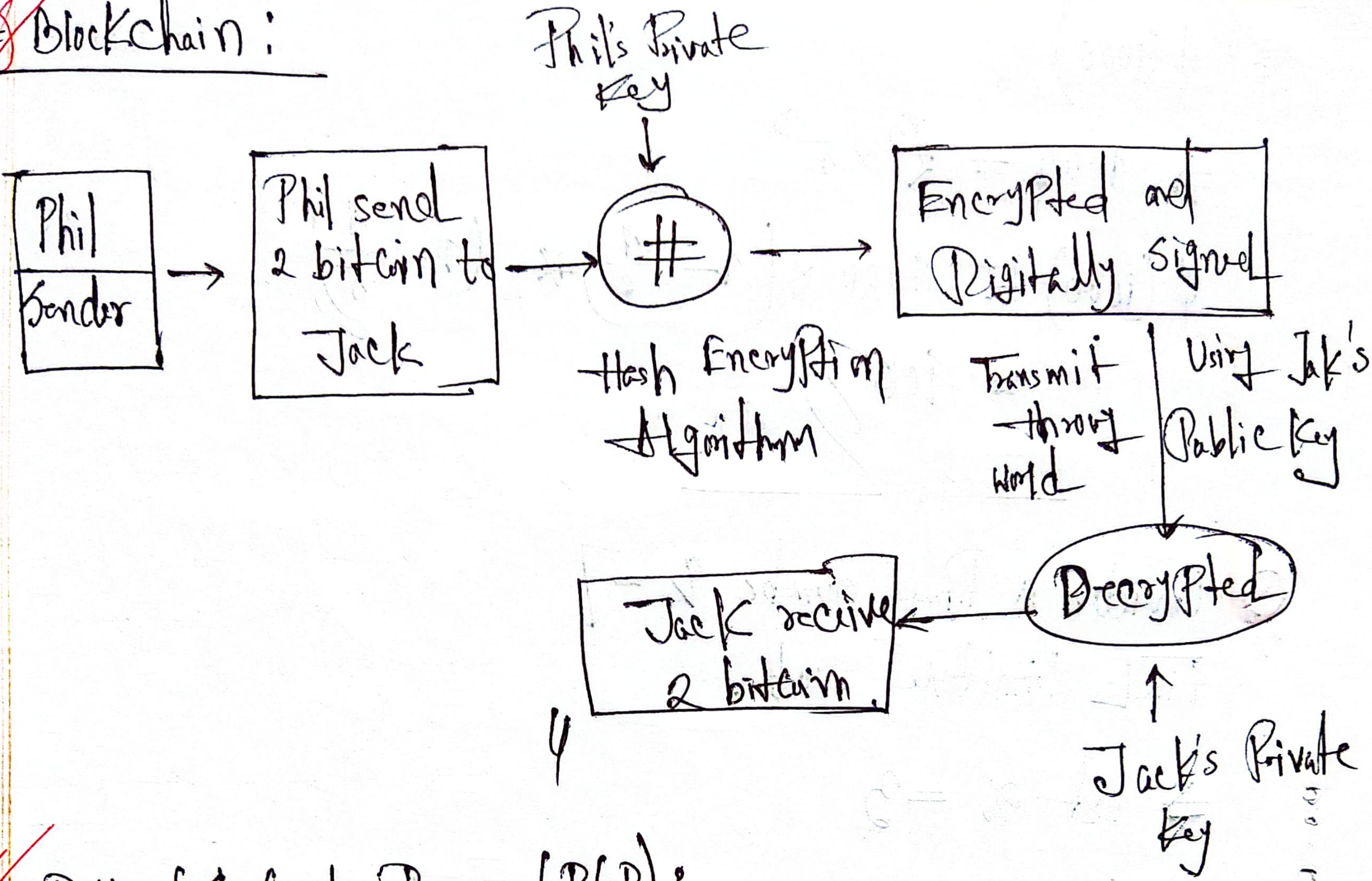
(ii) Propagation " [Virus is fruitful and multiplies]

(iii) Triggering "

(iv) Execution "

Segment — 8

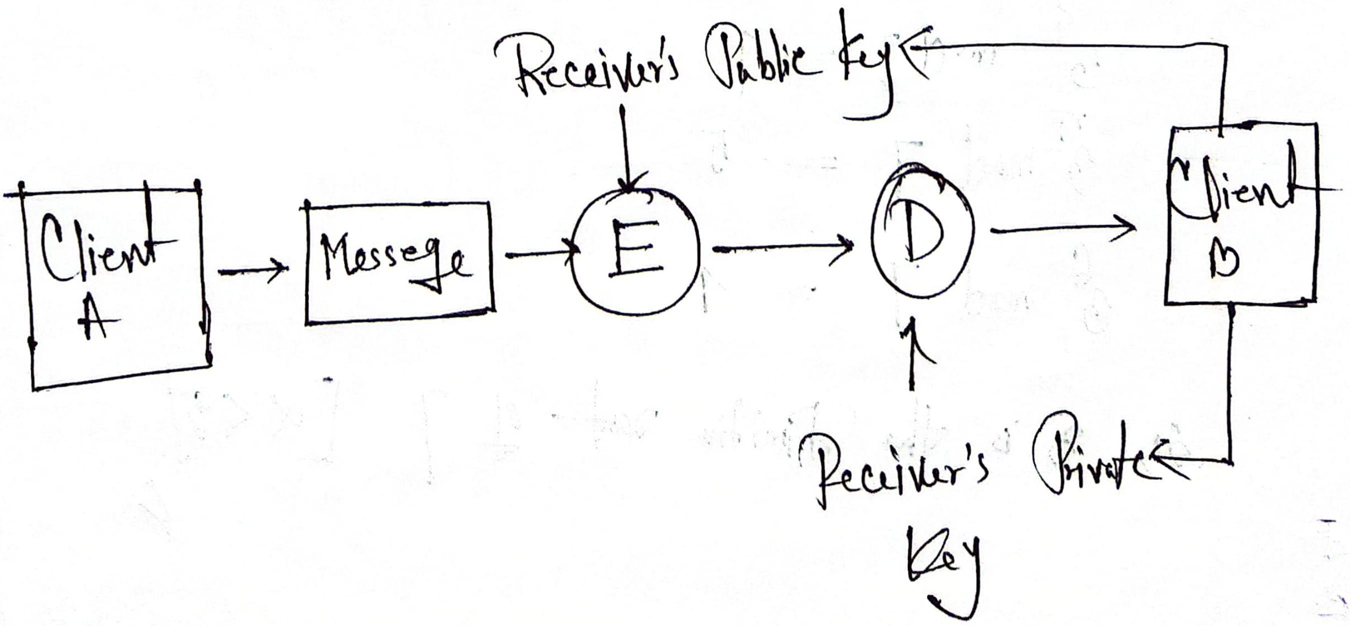
Blockchain:



Betty ~~God~~ Good Privacy (PGP):

⇒ Is an encryption system used for both sending emails and encrypting sensitive files

⇒ Kerberos and SSL encryption used by [PGP]



$\phi(1000)$

$\Rightarrow 1000 = 2^3 \times 5^3$

$\phi(1000) = 1000 \times \left(1 - \frac{1}{2}\right) \times \left(1 - \frac{1}{5}\right)$

$= 400$

Consider a Prime number $q = 7$

Find Primitive root.

$\Rightarrow$ Let $\alpha = 3$

Now,

$\alpha^1 \bmod 7 \Rightarrow 3^1 \bmod 7 = 3$

$3^2 \bmod 7 = 2$

$3^3 \bmod 7 = 6$

$3^4 \bmod 7 = 4$

$3^5 \bmod 7 = 5$

$3^6 \bmod 7 = 1$

So, 3 is the Primitive root of 7 $[\alpha < q]$

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